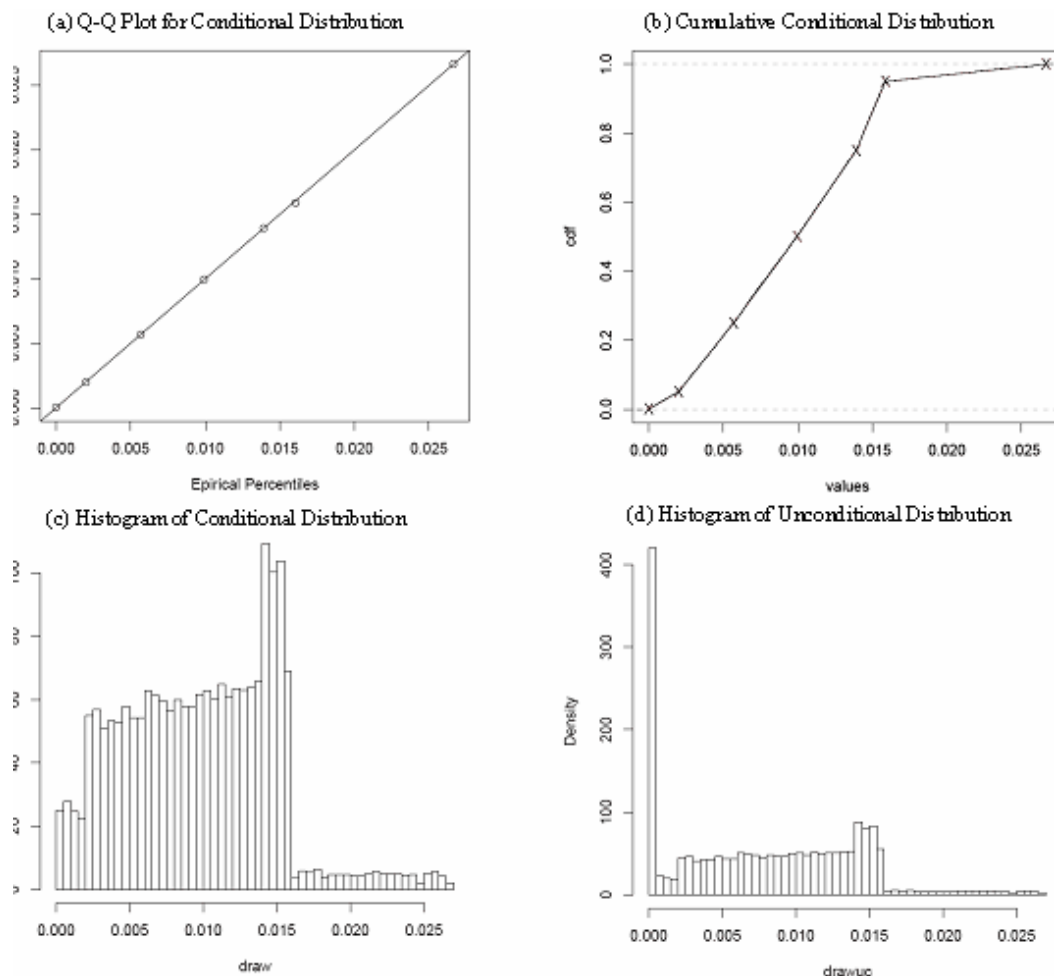


G.1.1.1.4.12 Expert L

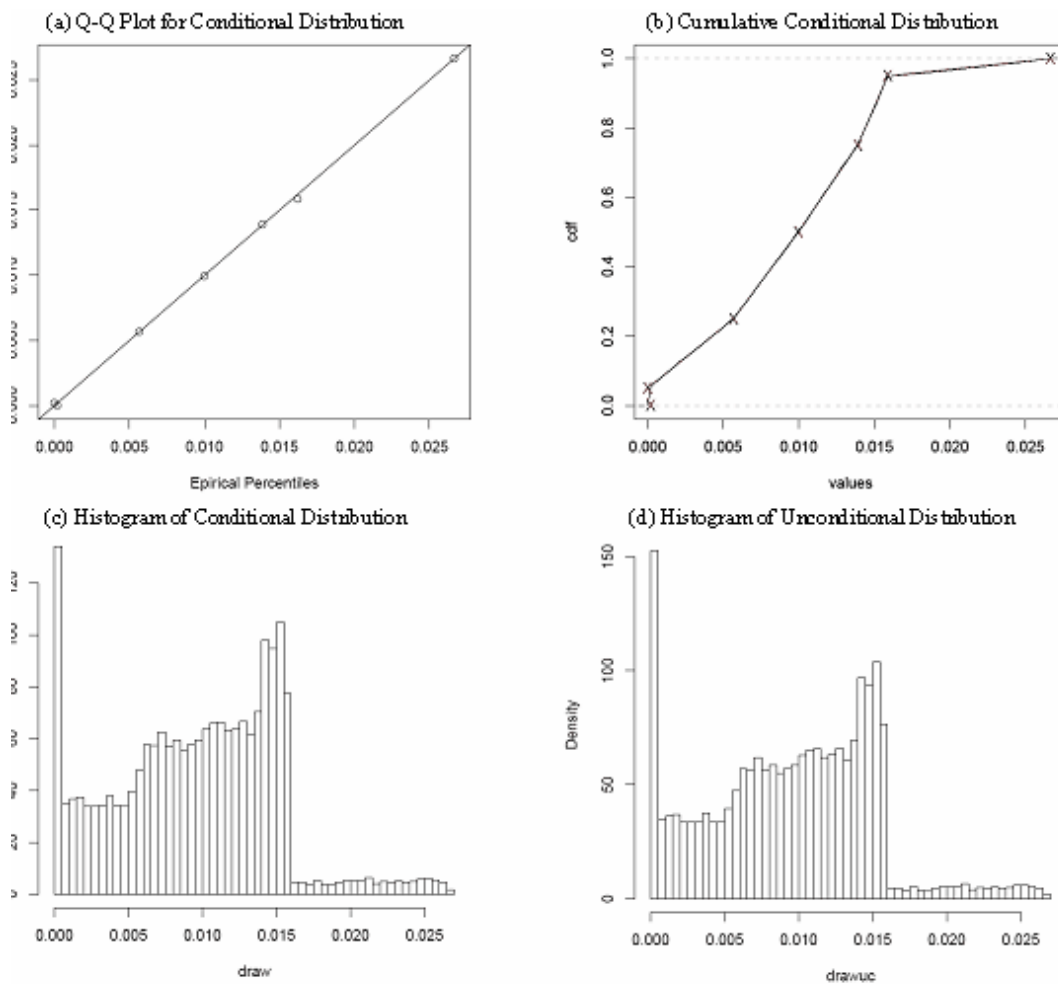
Figure G-12. Characteristics of the Random Draw from the Approximated Distribution of the PM_{2.5} Effect Specified by Expert L
(1) Results for the range 4-10 µg/m³

**Notes:**

Expert L specified a non-parametric distribution using five percentile points. We linearly interpolated the CDF between them. Panel (a) shows q-q plot of the expert percentiles and empirical percentiles for the draw. Panel (b) shows empirical CDF associated with the draw, the red "X" marks indicate corresponding expert percentiles. The distribution was conditional on causality. We created a corresponding unconditional distribution by adding extra 25 percent zeros to the draw. Panels (c) and (d) show the respective distributions.

The experts specified distributions for the percent changes in the relative risk. The distribution of the corresponding PM_{2.5} effects was the following transformation of the percent change in relative risk Z - $\log(1+(Z/100))$.

Figure G-12. Characteristics of the Random Draw from the Approximated Distribution of the PM_{2.5} Effect Specified by Expert L (continued)
(2) Results for the range >10-30 µg/m³



Notes:

Expert L specified a non-parametric distribution using five percentile points. We linearly interpolated the CDF between them. Panel (a) shows q-q plot of the expert percentiles and empirical percentiles for the draw. Panel (b) shows empirical CDF associated with the draw, the red "X" marks indicate corresponding expert percentiles. The distribution was conditional on causality. We created a corresponding unconditional distribution by adding extra 1 percent zeros to the draw. Panels (c) and (d) show the respective distributions.

The experts specified distributions for the percent changes in the relative risk. The distribution of the corresponding PM_{2.5} effects was the following transformation of the percent change in relative risk Z - $\log(1+(Z/100))$.

G.1.1.2 Baxter et al. (2017)

Baxter et al. (2017) used a cluster-based approach to evaluate the impact of residential infiltration factors on intercity heterogeneity in short-term PM_{2.5}-mortality associations. The authors performed a city-level time series assessment using mortality data from the National Center for Health Statistics (NCHS). They included 77 U.S. cities in their analysis of 24-hour non-accidental mortality across all ages. PM_{2.5} exposure estimates were based on average daily monitored PM_{2.5} concentrations in each city and the 2-day moving average (lag 0-1 days) of PM_{2.5} concentration was included in the model. The authors utilized Poisson regression models at the city-level and then a meta-regression model to aggregate the city-specific effect estimates into an overall effect estimate. Specifically, in the first stage, the authors ran single-city Poisson time-series models adjusting for temperature and dew point temperature and including variables for previous day temperature, temporal trends, and trends by age. The second stage was a meta-regression with cluster analysis (5 clusters) based on characteristics of residential infiltration.

Short-term Mortality, Non-Accidental

In a single-pollutant model, the coefficient and standard error are estimated from the percent increase in 24-hour non-accidental mortality (0.33%) and 95% confidence intervals (95% CI: 0.13-0.53%) for a 10 µg/m³ increase in daily 24-hour mean PM_{2.5} for a 2-day moving average (lag 0-1 days) (Baxter, et al., 2017, in results section text).

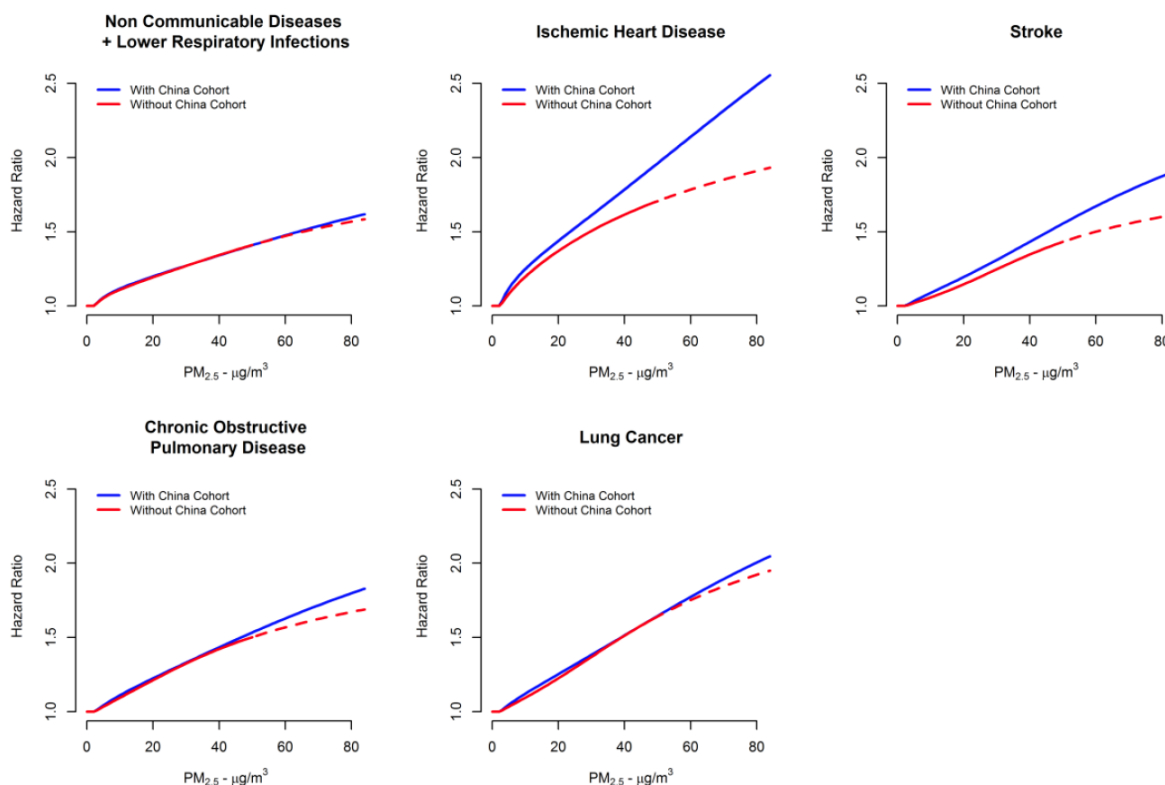
G.1.1.3 Global Exposure Mortality Model (Burnett et al., 2018)

The Global Exposure Mortality Model (GEMM) is a family of functions that can be used to estimate the global burden of disease attributable to ambient fine particle (PM_{2.5}) exposure over the entire global exposure range. The GEMM is the latest iteration of earlier models with the same purpose, such as the Integrated Exposure Response (IER) models (Burnett et al., 2014). The GEMM improves upon these models by excluding studies of non-ambient particle exposure, identifying a single counterfactual concentration (2.4µg/m³), and quantifying all non-accidental mortality related to ambient PM_{2.5} concentrations relative to that level. The GEMM was constructed by combining data from 41 cohort studies in 16 countries that examined the relationship between ambient PM_{2.5} concentrations and mortality. These studies spanned a PM_{2.5} range of 2.4µg/m³ to 83.7µg/m³, which encompasses the 2015 population weighted outdoor exposure levels for 97% of the global population.

The GEMM consists of 83 non-linear relative risk functions that relate changes in ambient PM_{2.5} concentrations to changes in six different mortality endpoints: non-communicable diseases plus lower respiratory infections (NCD+LRI), cerebrovascular disease (CEV), chronic obstructive pulmonary disease (COPD), ischemic heart disease (IHD), lung cancer (LC), and lower respiratory infections (LRI). The functions are applicable to adults age 25-99, and age-specific functions are also provided for NCD+LRI, CEV, and IHD because cardiovascular risk factors, including PM_{2.5}, are known to decline with age. The authors also prepared one set of functions that exclude the

Chinese male cohort study because it had the largest $\text{PM}_{2.5}$ range ($15.4 \mu\text{g}/\text{m}^3$ to $83.7 \mu\text{g}/\text{m}^3$). The two sets of functions do not differ significantly for NCD+LRI, COPD, and LC; however, excluding the Chinese male cohort yielded lower hazard ratio predictions for CEV and IHD (Figure G-13).

Figure G-13. Figure S6 from Burnett et al., 2018 depicting GEMM hazard ratio predictions for NCD+LRI, IHD, CEV (stroke), COPD, and LC with (blue line) and without Chinese Male Cohort (red line).



Within BenMAP-CE, each GEMM relative risk function is applied to an air quality scenario by first using the applicable GEMM equation to estimate the log-linear beta coefficient for the $\text{PM}_{2.5}$ change in each grid cell, and then inserting that beta value into the standard BenMAP-CE formula $(1 - (1/\text{EXP}(\text{Beta} \cdot \text{DELTAQ}))) \cdot \text{Incidence} \cdot \text{POP}$. Additional function logic prevents the GEMM from predicting health benefits below the counterfactual concentration of $2.4 \mu\text{g}/\text{m}^3$. As a result of this linear approximation approach, the calculated BenMAP point estimate(s) are not a perfect reflection of the non-linear GEMM functions. However, this approach greatly reduces the implementation complexity of these functions while maintaining a reasonable level of accuracy, typically within 10% of the non-linear function.